



Multi-Faceted and Network-Based Hybrid Resilience Assessment on Rural Shrinkage in Germany and Türkiye

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Abstract: Despite advances, spatial resilience planning remains constrained in its integration of complex system principles to address slow-variable disturbances. This study provided a methodological test of a novel multi-faceted and network-based hybrid resilience assessment that examined rural shrinkage in paired regions of Germany (Lüneburg) and Türkiye (Trakya). The method integrated Specified Resilience Assessment (SRA) and General Resilience Assessment (GRA) under Socio-Ecological Systems (SES) and Complex Adaptive Systems (CAS) lenses and operated through five steps. SRA employed (i) a multi-faceted survey to identify prioritized factors, solutions, and institutional roles/success and (ii) Relational Network Analysis (RNA) to assess complex factors and leverage points; GRA computed (iii) Spatial Network Analysis (SNA) to identify physical connectivity as hubs and sub-clusters; (iv) correlation analysis to determine significant variables among socio-demographic, land-use, facility, and network variables, and (v) *k*-means clustering to map shrinkage urgency levels. The synthesized outputs generated two operational strategies: strengthening sub-centers and connecting shrinking settlements to these hubs. While the strategies of Germany focused on the needs of the elderly and innovative digital solutions ($wd \approx 28$), examples of Türkiye emphasized ecological concerns and the support of cooperatives as a leverage ($wd = 54$). GRA highlighted weighted degree (up to $r = 0.79$) and urban-industrial land cover ($r \approx 0.6$) as critical drivers of stability; meanwhile, distance to the center ($r \approx -0.55$) significantly correlated with shrinkage. Despite limitations of sample size and manual network construction, the study operationalized SES/CAS concepts for slow variables and integrated both qualitative and quantitative insights. It advances resilience research in sustainable spatial development by demonstrating a proof-of-concept and transferable decision-support workflow, while scaling and automation point to the directions for future research.

Keywords: Complex adaptive systems; Socio-ecological systems; Specified Resilience Assessment; General Resilience Assessment; Network analysis; Rural development; Strategic spatial planning; Urban-rural linkages

1. Introduction

Advances in complexity and resilience theories offer novel and pivotal methodologies for strategic spatial planning. Complexity theory, as a metatheory (Craig, 2020), provides integrated, relational, and network-based perspectives for analyzing complex systems, while resilience research examines Socio-Ecological Systems (SES) through the principles of Complex Adaptive Systems (CAS) to understand the transition to sustainability (Preiser et al., 2018; Sellberg et al., 2021). These approaches yield valuable insights into spatial planning by investigating dynamic system behaviors and responses to wanted/unwanted changes, such as external shocks or internal slow variables (Peter & Swilling, 2014; Reyers et al., 2022; Sellberg et al., 2021; Wilkinson, 2012).

Despite these theoretical and methodological advances, resilience practices in spatial planning and policy remain fragmented, and limitations exist in adapting CAS and SES approaches. Existing studies often relied on i) either qualitative or quantitative methods; ii) targeted a single scale or aspect; or iii) prioritized external stressors, notably the impact of climate change on urban systems (Dianat et al., 2022; Kong et al., 2022; Nel et al., 2018; Shi et al., 2021; Trump et al., 2018). This leaves a critical gap for a comprehensive hybrid resilience assessment method that can address slow variables within spatial planning (Dianat et al., 2022; Wang et al., 2018). Similarly,

within international resilience policy, detailed urban infrastructure resilience assessment frameworks and disaster-focused toolkits are now common. Recent reports have incorporated multi-level governance, multi-temporal assessment, and cascading multi-risk analyses with urgency levels (CDRI, 2025; UNDRR, 2015; UNDRR & CDRI, 2023). Yet there is still a lack of comprehensive CAS frameworks that could address socio-economic slow-variable factors and support their translation into spatial planning applications at the micro level.

To bridge these academic and policy gaps, this study develops a multi-faceted and network-based hybrid resilience assessment method to integrate Specified Resilience Assessment (SRA) and General Resilience Assessment (GRA) under SES and CAS perspectives (Craig, 2020; Zellner & Campbell, 2015). The method is empirically tested on rural shrinkage in paired regions of Germany and Türkiye, which have contrasting development contexts yet comparable territorial structures. Rural shrinkage is selected as a representative slow-variable, due to its systemic impacts on sustainable development and close linkages with the trends of urbanization (Jarzebski et al., 2021; Li et al., 2019). It is driven by local factors such as rural-urban interactions, out-migration, counter-urbanization, and demographic changes; however, at the regional scale, it is a symptom of more complex processes as part of regional/territorial SES (Bai et al., 2016; ESPON, 2017). Through feedback loops and cascading effects, the consequences of rural shrinkage can disrupt the equilibrium and functionality of connected and wider SES.

Accordingly, the research question guiding this study is: *How can a multi-faceted and network-based hybrid resilience assessment method inform spatial planning interventions that strengthen resilience against rural shrinkage?* The methodology consists of two components and five steps. Within the SRA component, i) the multi-faceted survey gathers qualitative insights such as place-based risks, solutions, institutional roles, and success; ii) Relational Network Analysis (RNA) examines the interdependencies among complex factors and leverage points. Within the GRA component, quantitative approaches are employed; iii) Spatial Network Analysis (SNA) is used to assess spatial connectivity patterns; iv) correlation analysis is used to identify significant variables, and v) *k*-means clustering is used to distinguish settlement vulnerabilities. Finally, synthesizing these outputs reveals actionable strategies that guide place-based investments by strengthening sub-centers and linking vulnerable areas. The study develops a policy-aligned and replicable model that translates high-level strategies into micro-level applications, consistent with the United Nations Human Settlements Programme (UN-HABITAT, 2017), the European Observation Network for Territorial Development and Cohesion (ESPON, 2017; ESPON, 2021), and the Organisation for Economic Co-operation and Development (OECD, 2025) which call for place-based, data-driven, integrated, and networked urban-rural linkages, with multi-level governance, innovative service delivery, and smart-shrinkage strategies to address rural shrinkage.

The method advances spatial resilience planning in five main ways; it (i) operationalizes SES/CAS perspectives through a multi-faceted and network-based method that reveals relational interdependencies and spatial linkages; (ii) combines qualitative SRA-RNA with quantitative GRA-SNA, in a single hybrid resilience assessment; (iii) extends resilience assessment to a slow-variable disturbance by testing the approach on rural shrinkage; (iv) bridges strategic policy goals and operational actions by deriving data-driven selection rules for sub-centers and linkages at the settlement scale; (v) offers a transferable and standardized workflow that can be adapted to other issues and outlines pathways for embedding SRA and GRA outputs into spatial planning practices. Together, these contributions provide planners and policymakers with a network-based decision-support approach for guiding resilient and sustainable spatial development in the face of complex challenges.

2. Conceptual Foundations: Resilience Assessment and Network Analysis

This section sets the conceptual background of the proposed methodology. Subsection 2.1 reviews resilience assessment in spatial planning and its limitations; 2.2 outlines specified and general resilience within SES/CAS perspectives; 2.3 introduces network theory and RNA-SNA as a quantitative bridge.

2.1 Resilience Assessment in Spatial Planning

Resilience assessment in spatial planning and policy has advanced through various scales and domains, from regional economic strategy and mitigation of urban hazards to rural community viability, yet remains constrained by fragmented definitions, reliance on either qualitative or quantitative methods, and limited adoption of complex systems perspectives like multi-faceted comprehensive assessments.

Studies of regional resilience often adopt CAS perspectives, viewing regions as open and dynamic systems shaped by multi-level flows of production, population, and employment (Holling & Goldberg, 1971; Peng et al., 2017). These works highlight adaptive governance and dynamic monitoring to understand feedback loops and non-linear dynamics (Hartman & De Roo, 2013; Holling, 2001; Palekiene et al., 2015; Pendall et al., 2009), thereby supporting dynamic analysis of institutional, causal and spatial networks. However, they often overlook multi-scale perspectives that connect local conditions with regional contexts.

Research on urban resilience largely targets rapid shocks and external disturbances (e.g., impacts of climate

change and floods) (Kong et al., 2022; Spaans & Waterhout, 2017). At the level of international policy, disaster resilience assessment frameworks provide metrics, toolkits, and comparative scorecards, to emphasize multi-level governance, multi-temporal monitoring, and cascading multi-hazard risks, primarily for critical urban infrastructure systems (CDRI, 2025; UNDRR, 2015; UNDRR & CDRI, 2023). In parallel, urban resilience research uses computational tools including agent-based modelling, cellular automata, and network (graph) theory to assess the robustness of infrastructure (Boeing, 2018; Espada et al., 2015; Marcus & Colding, 2014; Roggema et al., 2012). Additionally, adaptive governance, soft-system management (Crowe et al., 2016; De Luca et al., 2021) as well as the examination of CAS attributes such as diversity, adaptability, cooperation, connectivity, modularity, robustness, abundance, and flexibility have been applied (Ahern, 2011; Eraydın & Taşan-Kök, 2013). However, both policy toolkits and academic studies still rely on either qualitative (Dianat et al., 2022; Faulkner et al., 2020) or quantitative methods in isolation (Cardoso et al., 2020; Shi et al., 2021). This situation highlights the need for integrated and hybrid methods in urban resilience assessment (Kong et al., 2022; Wang et al., 2018), together with the need to address slow variables such as socio-economic changes (Eraydın & Özatağan, 2021).

Research on rural resilience focuses on maintaining essential functions such as agriculture, rural services, and nature conservation (Heijman et al., 2019). Theoretical research encompasses SES holistic concepts to explore social, economic, environmental, and governance interdependencies (Akgün et al., 2014; Chirisa & Nel, 2022; Schouten et al., 2009; Scott, 2013). Qualitative research focuses on building community resilience (Cox & Hamlen, 2015; UNDP, 2017) and innovative interventions at the local level (Roberts et al., 2017; Roberts et al., 2015; Steiner & Atterton, 2015). Quantitative methods use diversity indexes (Cox & Hamlen, 2015; Quaranta & Salvia, 2014), spatio-temporal dynamics, and scenario analysis against socio-economic challenges (Colantoni et al., 2020; Sánchez-Zamora et al., 2014; Zhou & Hou, 2021). Recent studies have addressed rural shrinkage (Li, 2023), highlighted the network resilience (Yu et al., 2024), and employed multi-criteria approaches to evaluate social, economic, ecosystems, and adaptive management capacities (Allen et al., 2016). Furthermore, for resilient rural areas, factors conducive to tangible and intangible success are identified, leading to the survey design in this study. Tangible factors include the diversity and accessibility of resources (Quaranta & Salvia, 2014), whereas intangible factors include social capital, institutional networks, strong community ties, and effective governance (Anthopoulos et al., 2017; McManus et al., 2012). Moreover, leadership, networks, and trust are recognized as key factors of resilience in the social systems, facilitating information flow, collaboration, and effective knowledge exchange (Morrison, 2014).

Although advanced and diverse methods exist across regional, urban, and rural contexts, resilience assessment remains constrained by ambiguous definitions, separated qualitative and quantitative practices, and limited adoption of complex systems perspectives (Linkov & Trump, 2019; Pendall et al., 2009; Schipper & Langston, 2015; Zhu et al., 2025). These gaps continue to hinder comprehensive and multi-faceted resilience assessment in spatial planning. Advancing the field therefore requires adapting a complex systems perspective for clearer conceptualization, robust datasets, hybrid and multi-faceted methods capable of tracking complex disturbances, interventions and relations across multiple scales, dimensions, governance levels, and time frames (Kinzig et al., 2006; Pendall et al., 2009; Schouten et al., 2009; Skerratt, 2013).

2.2 Resilience Assessment in Socio-Ecological Systems

SES studies address resilience as a CAS attribute and provide key concepts/methods for applied resilience and sustainability transitions (Sellberg et al., 2015). While SES approaches align closely with the complexity of spatial planning, their integration into planning practice remains limited, particularly in the development of quantitative methods to assess resilience attributes (Malmberg et al., 2022). Most SES applications rely on qualitative methods such as questionnaires, in-depth interviews, and public meetings (Lee et al., 2020; UNDP, 2017) that could capture local knowledge yet lack spatial context and quantitative reach.

To guide this study, the five-step operational framework of the ResilienceAlliance (2010) is adopted. This framework involves: system definition (boundaries, function, and elements), model change, and thresholds to reveal system dynamics, analyze cross-scale interactions (cascading effects and feedback loops), explore system adaptive management, finally assess resilience, synthesize findings, and establish resilience-oriented governance.

This framework (ResilienceAlliance, 2010) further provides a detailed method for Specified and General Resilience Assessments, which structures the hybrid methodology developed in this study. SRA examines particular conditions and risks of a given system through five guiding questions, which form the foundation of multi-faceted matrix used in this study (Section 3.2). The first question is (i) Resilience of what and to whom defined the system's core functions, boundaries, elements, sub-components, and linkages with upper and lower systems; (ii) Resilience to what classifies disturbances by origin (ecologic, economic, social, technological, and political) (Müller, 2011) and by temporal features: slow variables as internal factors impacting wider zones like environmental, socio-economic, and demographic changes (Folke et al., 2004; Walker et al., 2006), or fast variables as external shocks like earthquakes, floods, and diseases (Spaans & Waterhout, 2017). Fast variables are often more noticeable and more examined, whereas recognizing and monitoring slow variables is crucial to ensure

the stability of the system in the long term due to their broader impacts (Walker et al., 2012). In this regard, rural shrinkage, a slow variable with systemic effects, is selected as the empirical test case; (iii) System dynamics examines the adaptive-cycle phases (exploitation, conservation, release, and reorganization), while each phase displays distinct resilience levels and paces of system change shaped by the accumulations and levels of connectedness in the system (Gunderson & Holling, 2002). Network theory can address gaps in quantifying accumulation and connectedness through dynamic monitoring; (iv) System governance analyzes multi-level governance interrelations among macro, mezzo, and micro levels (Folke et al., 2011), where RNA can analyze these interrelations; and (v) System capacities aims to develop transition, adaptation, and coping capacities, and to identify measures (policies, strategies, and projects) aligned with predefined risks/ intensity of disturbances (high, moderate, and low) across multi-scale (macro, mezzo, and micro) and multi-dimensional contexts (social, ecological, and economic), multi-level governance (e.g., governmental, regional, and local), and multi-temporal phases (long, mid, and short) (Anderies et al., 2013; Béné et al., 2015).

GRA explores common attributes of complex systems to absorb the disturbances, including diversity, connectedness, flexibility, robustness, openness, resource availability, and modularity. Although quantitative methods for assessing CAS attributes are still evolving, network analysis serves as a useful tool in operationalizing many of them, particularly diversity, connectivity, and modularity, which are naturally linked to system definition and dynamics. This body of work also highlighted the importance of analyzing feedback loops and cascading effects to understand internal regulatory mechanisms and non-linear dynamics of systems (Carvalhoes et al., 2021; Sellberg et al., 2021), which highlights contribution of RNA.

Collectively, resilience in SES studies often employ multi-faceted, relational, and networked approaches that recognize CAS attributes. However, spatial analysis and their operationalization remain limited and challenging (Baird et al., 2016). Embedding these characteristics in spatial resilience planning underpins the multi-faceted and network-based hybrid methodology in this study, to enable a compact resilience assessment tool for both SES and planning domains.

2.3 Network Analysis—A Quantitative Bridge from Concept to Practice

Network (graph) theory provides a methodological bridge for translating spatial, functional, causal, and relational linkages among places, actors, resources, and risk factor-solution pairs into a comparable quantitative model (Estrada & Knight, 2015). Since the 1990s, it has been pivotal in analyzing CAS (Barabási, 2016), representing system elements as nodes and relationships as edges with potential weights and directions. An adjacency matrix records whether any two nodes are connected (1) or not (0) (Estrada & Knight, 2015). Algorithms then extract two types of network metrics. Local metrics calculate specific attributes of an entity, such as weighted node degrees (highly connected nodes) for identifying key intervention hubs or leverage points. Global metrics such as network topology, centrality, and modularity, reveal the overall system structure (centralized or distributed) and identify sub-clusters that are critical for risk assessment and management (Hernandez & Miegheem, 2011).

In spatial planning, network analysis has been applied to institutional and social systems by Actor–Network Theory (Bassi et al., 2014; Castellani & Hafferty, 2009), and to physical networks like transportation, infrastructure (Yazdani et al., 2011) as well as geographical and ecological networks (Zetterberg et al., 2010). Across these studies, network connectivity, i.e., SES's connectedness attribute, is recognized as a critical determinant of resilience across institutional and spatial contexts (Müller, 2011; Peng et al., 2017). This aligns with international policy priorities that emphasize strengthening urban-rural linkages to foster regional resilience and sustainable development (ESPON, 2017; UN-HABITAT, 2017). RNA extends this potential by framing causal interactions with a multi-faceted approach. To take an early example in Panarchy (Gunderson & Holling, 2002), rangelands were described as complex ecological-economic systems and human-adaptable factors were mapped across multi-level governance, multi-scales (national, regional, and local) and multi-dimensions (social, economic, and ecological) (Gunderson & Holling, 2002). While such diagrams conceptualized multi-faceted interrelations among factors, they lacked quantitative assessment of factor solution pairs. Other decision-support methods such as analytical hierarchy process and analytical network process assessed pairwise criteria (Kong et al., 2022), Decision Making and Trial Evaluation Laboratory (DEMATEL) (Parizi et al., 2021) explored dependencies among criteria, and system dynamics modelled the feedback loops (Feofilovs et al., 2020). These methods offer valuable insights yet remain limited in quantifying interdependencies between risk and solutions, thus highlighting the need for further development (Bai et al., 2016).

This study responds to these gaps by applying network analysis in two complementary roles within spatial resilience assessment by engaging a multi-faceted approach. First, (i) relational role: treating factors and solutions as nodes to assess causal relations, identifying complex risk factors, and detecting leverage points for specified resilience. Second, (ii) spatial role: quantifying the physical connectivity of settlements and identifying sub-centers and sub-clusters to assess general resilience attributes.

Conceptual networks are structured by defining nodes as the elements of planning systems like places, institutions, risk factors or solutions, and edges as the functional or relational linkages among them, such as

material flow, information exchanges, institutional ties, or causal relationships (Figure 1). These linkages are organized across multiple dimensions (social, economic, and ecological), scales (macro, mezzo, and micro) and governance levels (regional, provincial, and local) (Fraccascia et al., 2018; Linkov & Trump, 2019; Neal, 2013; Perez et al., 2016; Yu et al., 2024).

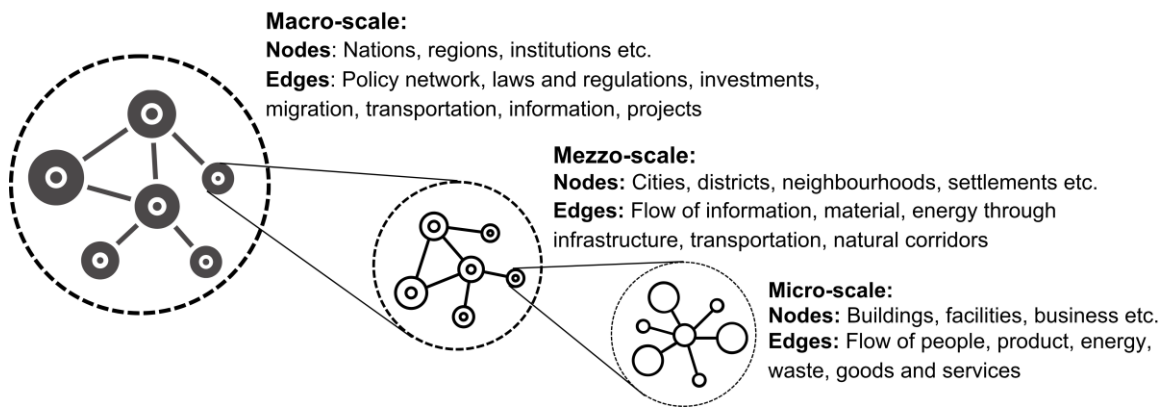


Figure 1. Conceptualization of the nodes and edges of the planning system as a multi-faceted network

Overall, this multi-faceted relational/spatial network framework offers planners a data-driven tool to analyze complex planning challenges, address complexity and resilience attributes, and support strategic interventions. It forms the methodological basis of the study, which is operationalized in the next section through Specified and General Resilience Assessments.

3. Materials and Methods

Figure 2 illustrates the five analytical steps of the resilience assessment method. SRA encompasses qualitative and participatory tools, including (i) a multi-faceted and semi-structured survey; and (ii) RNA. GRA employs quantitative spatial analysis, including (iii) spatial network; (iv) correlation; and (v) *k*-means analyses, using various socio-demographic and geospatial data. The outputs of SRA and GRA are then synthesized to produce place-based strategic interventions. The detailed inputs/outputs and analytic checks for each operation are provided in Supplementary File 1–11, including RNA sensitivity (Supplementary File 6), SNA robustness checks (Supplementary File 8), Moran's *I* diagnostics (Supplementary File 9), and *k*-means centroids and stability tests (Supplementary File 10–11). Complete correlation matrices and False Discovery Rate (FDR)-adjusted *q*-values, along with other supporting materials, are available in the repository (see Data and Code Availability).

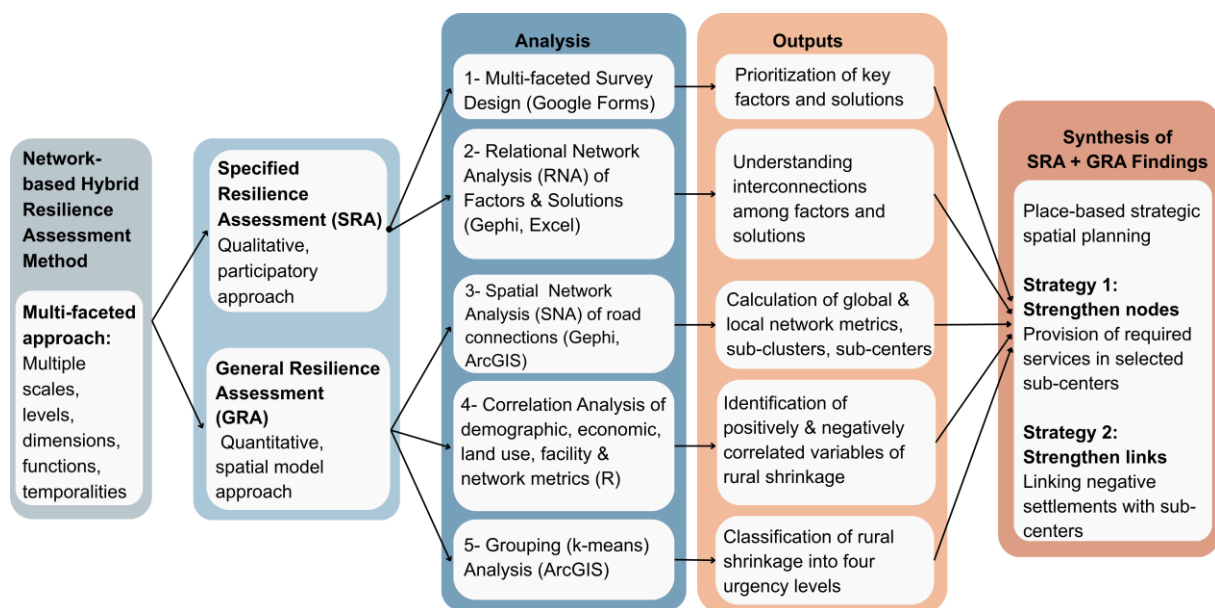


Figure 2. Illustration of the research design and steps

To compare resilience dynamics across the contexts of developed and developing countries while holding background conditions as similar as possible, paired regions and districts were selected in Germany and Türkiye. Selection criteria included population size, land area, population density, share of elder population (65+), distance to metropolitan centers, proximity to ports, and railway connectivity (comparative case-selection tables: Supplementary File 1). The multi-scale case study design involved three nested scales (Figure 3).

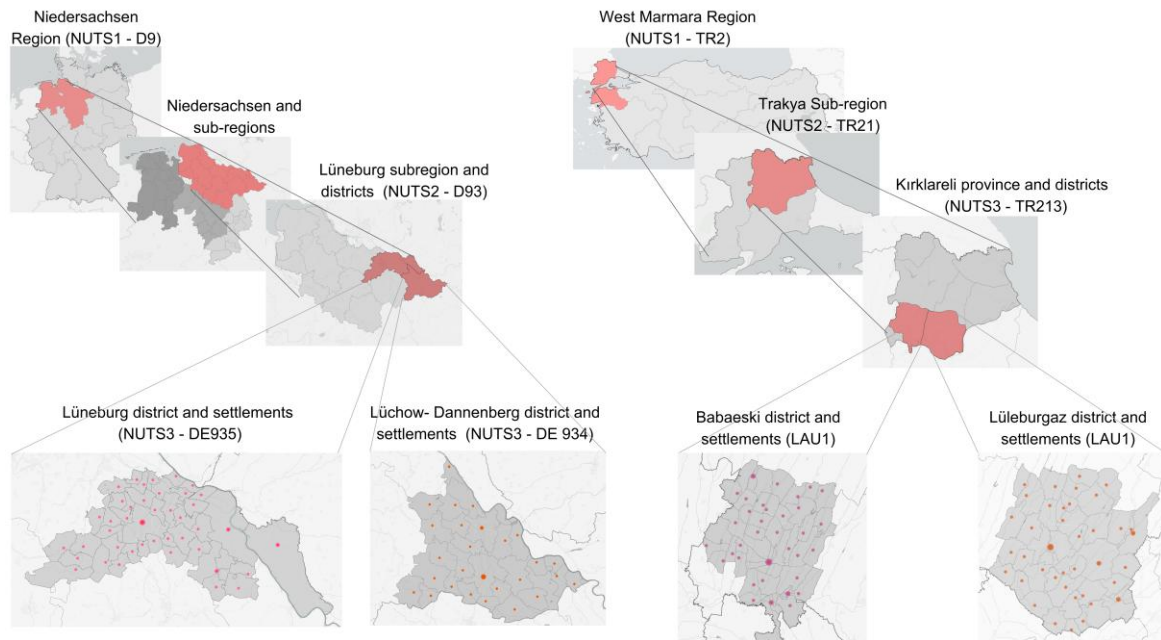


Figure 3. Case study areas in Germany and Türkiye

At the macro-scale (NUTS2 level), Lüneburg (DE93) and Trakya (TR21) regions were compared based on their comparable area and population size. At the mezzo-scale and within each region, neighboring districts were paired; while one was experiencing shrinkage (negative population change), the other was not (positive population change). Due to different provincial division of countries, NUTS3 level DE935 Lüneburg (positive) and DE934 Lüchow-Dannenberg (negative) counties from Germany were compared with LAU1 districts Lüleburgaz (positive) and Babaeski (negative) from Türkiye. At the micro-scale, respective villages and neighborhoods of each case were analyzed.

3.1 Specified Resilience Assessment Method

A multi-faceted matrix (Figure 4) was designed as a compact and structured SRA framework for complex planning systems. The columns reflect the five specified resilience questions (see Section 2.2). The rows facilitate a complex system analysis across multi-scale (macro, mezzo, and micro); multi-dimensional (social, economic, and ecological); and multi-temporal phases (long, mid, and short terms).

The resulting grid supports planners to comprehensively describe the structure, potential risks, institutional roles, and strategic interventions proposed by the system to understand and improve system resilience. In brief, the matrix prompted five questions: (1) define system boundaries, elements, sub-components, and core functions; (2) classify risk factors by type, pace, and spatial impact; (3) distinguish adaptive cycle phases, requiring a dynamic assessment of connectivity and accumulation to be consistent with general resilience; (4) identify roles of relevant institutions; and (5) build system capacity to mitigate risks through multi-scale, multi-dimensional, and multi-temporal measures, policies, plans, and actions.

Based on this multi-faceted matrix, an online semi-structured survey was designed and applied in case studies in 2019. As a methodological test, SRA was explicitly exploratory rather than based on probability sampling; its purpose was to identify and prioritize place-based risks, solutions, and organizational roles. No population-level generalizations were claimed. A purposive and expert-oriented multi-stakeholder sampling strategy was adopted in the survey. Invitations were sent to 217 contacts in Germany and 84 in Türkiye; these were assembled from official provincial/municipal directories, development agencies, universities, cooperatives/NGOs, and local business associations (survey invitation and response rates: Supplementary File 2). A total of 45 responses ($DE = 25$, $TR = 20$), corresponding to completion rates of 11% and 23%, were obtained respectively. German participants ($n = 25$) were all experts and mostly aged between 45–64. Turkish participants ($n = 20$) were equally divided between experts and workers and mostly aged between 25–44.

The survey comprised 6 sections with 36 questions, mixing open-ended questions with multiple-choice lists of factors and solutions derived from the literature (Section 2.1) (full survey form: Supplementary File 3). The importance of factors and solutions was rated on a 0–4 Likert scale and the performance of institutions on a 1–10 scale. On the scale of 0–4, scores above 3 were considered to be top priorities, between 2.5–3 were secondary, while those below 2.5 were excluded. Cronbach’s alpha coefficient was calculated as a diagnostic summary of internal consistency for the grouped survey items, factors and solutions, and for the social, economic, and ecological subgroups within factors. Alpha values were interpreted descriptively ($\approx 0.9^+$ excellent, 0.8–0.89 good, 0.7–0.79 acceptable). For Germany, Factors $\alpha = 0.933$ (excellent) and Solutions $\alpha = 0.804$ (good), whereas for Türkiye, Factors $\alpha = 0.865$ (good) and Solutions $\alpha = 0.875$ (good) (summary of Cronbach’s alpha results: Supplementary File 4).

A pilot RNA followed. The steps of the RNA algorithm are given in Figure 5. To create relational networks, case-level averages (0–4 scale) were calculated for each factor and solution based on the responses from survey. Items with an average ≥ 2.5 were retained. This dataset was enriched by the additional factors and solutions proposed by participants (additional responses: Supplementary File 5), which were assigned a default priority value of 3. Next, an undirected bipartite network was created between the retained factors and solutions. For each retained pair (f, s), an edge weight defined by the average of the two nodes was assigned. This weighted adjacency matrix was imported into Gephi 0.9.2 (Bastian et al, 2009) for calculations of the weighted degree.

Complex System Definition						
Multi-scale - Multi-dimensional - Multi-level - Multi-temporal framework		1-Resilience of what and whom?	2-Resilience to what?	3-System dynamics	4-System governance	5-System capacity
	Boundries of the system	What are the key components and priority issues ?	Which disturbances are changing the system?	In which adaptive phase is the system?	Which institutions are responsible ?	How can system capacities be developed?
	Macro-system (e.g., Region)	Upper system key components	Disturbances / Slow variables	Adaptive cycle phases	Macro Institutions	Transformative policies
	Social	Regional social components	Social variables	Phase 1,2,3,4	Regional social institutions	Social policies
	Economic	Regional economic components	Economic variables	Phase 1,2,3,4	Regional economic institutions	Economic policies
	Ecological	Regional ecological components	Ecological variables	Phase 1,2,3,4	Regional ecological institutions	Ecological policies
	Mezzo-system (e.g., provinces)	Focal system key components	Risks	Adaptive cycle phases	Mezzo Institutions	Adaptation strategies, plans
	Social	Province social components	Social risks	Phase 1,2,3,4	Provincial social institutions	Social plans
	Economic	Province economic components	Economic risks	Phase 1,2,3,4	Provincial economic institutions	Economic plans
	Ecological	Province ecological components	Ecological Risks	Phase 1,2,3,4	Provincial ecological institutions	Ecological plans
	Micro-system (e.g. district-villages)	Sub-system key components	Shocks	Adaptive cycle phases	Micro institutions, initiatives	Actions and projects to cope
	Social	District/village social components	Social shocks	Phase 1,2,3,4	District social services	Social projects
	Economic	District/village economic components	Economic shocks	Phase 1,2,3,4	District economic services	Economic projects
	Ecological	District/village ecological components	Ecological shocks	Phase 1,2,3,4	District ecological services	Ecological projects
Cross-scale interactions						

Figure 4. Multi-faceted matrix for Specified Resilience Assessment (SRA)

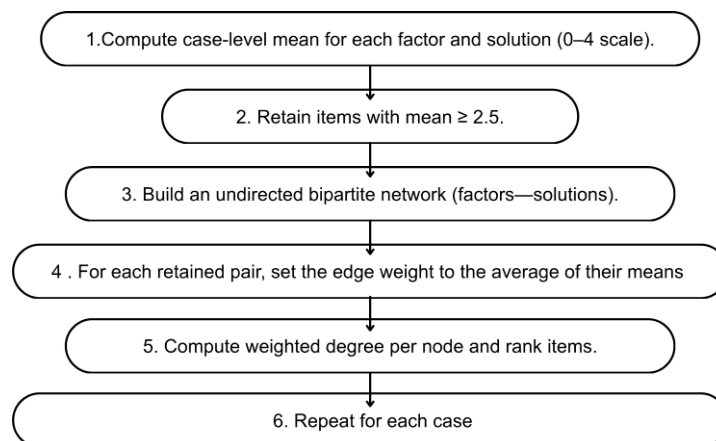


Figure 5. Relational Network Analysis (RNA) algorithm for factor-solution network

The Weighted Degree (*wd*) was selected as the primary centrality metric because it directly captured both the number and strength of links. In the bipartite network, high *wd* for factors indicates highly interconnected (complex) problem constellations, while high *wd* for solutions indicates potential leverage points, to align with the leverage concept in systems thinking (Meadows, 2008). Further, the sensitivity analyses of RNA rankings were tested using i) delete-*d* jackknife; and ii) a 5,000-draw bootstrap across cut-offs 2, 2.5, and 3 (a summary of RNA sensitivity analysis: Supplementary File 6). At 2, rankings were essentially unchanged across resamples; at 2.5, core priorities held with only minor shuffling around rank 6–8. At 3, top-rank ordering became unstable, identifying 2–2.5 as the defensible and respondent-robust range.

3.2 General Resilience Assessment Method

Data collection for the GRA method encompassed roughly 50 variables for each case. Data was gathered from multiple sources. For Germany, socio-demographic data were obtained from Eurostat and Destatis; for Türkiye, Turkish Statistical Institute (TUIK) and related ministries or municipalities. Land-cover data were obtained from the Coordination of Information on the Environment (CORINE) database (Corine, 2018) for both countries. Spatial data (administrative boundaries, roads, settlements, and point of interest (POIs)/facilities) were obtained from OpenStreetMap (OSM) (Ramm, 2022). Missing data for POIs was supplemented via Google Maps. Besides, distance to district center was calculated for each settlement. Land-cover data were re-aggregated into six classes: (1) urban; (2) industrial; (3) mine; (4) green urban areas and forests; (5) agricultural; and (6) wetlands and waterbodies. Land-cover shares were then calculated within each administrative boundary. POI data were regrouped into six facility categories counted for each case: (1) public; (2) commercial; (3) transportation; (4) recreation-attraction; (5) environmental; and (6) religious. Given the different data sources, administrative levels (NUTS3 for Germany vs. LAU1 for Türkiye), and statistical years, a multi-step harmonization process was applied. This included temporal alignment (e.g., selecting closest years/proxies), categorical alignment (e.g., unified re-classification), scale alignment (i.e., standardizing all variables to Z-scores) (detailed data harmonization table and categorization queries: Supplementary File 7).

To perform SNA, OSM hierarchical road data were categorized into 5 groups and assigned ordinal weights to serve as the basis for edge weights in the network. This 5-tier weighting system was chosen as it aligned with standard transport planning hierarchies and distinguished clearly between national/arterial and local/tertiary functions, capacity, and speed. Edge weights were assigned accordingly: railways (5), international motorway and trunk roads (4), primary national roads (3), secondary regional roads (2), and tertiary local roads (1). An adjacency matrix for settlement-to-settlement connections was then built in Excel. The weight of an edge between two settlements was calculated using a hybrid-cumulative approach to capture multi-modal capacity, prioritizing road and rail connections. The rules were as follows: in general, best road path was applied, if multiple separate road connections existed (e.g., a motorway and a primary road), the weight of the highest-ranking road was used. Also, if a single road segment was composed of different types (e.g., half primary and half secondary), its weight was summed proportionally. If a railway (5) existed in addition to any road connections, its value was summed with the best available road path. This matrix was imported into Gephi 0.9.2 (Bastian et al., 2009) to compute local and global network metrics: (i) local metrics (node degree and weighted node degree); and (ii) global metrics (modularity; various centralities-betweenness, closeness, and eigenvector) alongside clustering coefficient and number of triangles. Geo-layout was used for network visualization, later exported as shapefiles for adoption in Geographic Information Systems (GIS). SNA robustness analysis was conducted with two alternatives applied to every case: (i) binary adjacency (all positive links set to 1); and (ii) an exponential emphasis that disproportionately weighed high-capacity links (transforming each positive weight to $2^{(w-1)}$). Settlement rankings and top hub overlap to the baseline was compared using Spearman correlation and Jaccard similarity. Results remained closely aligned with the baseline (summary of SNA robustness results: Supplementary File 8).

Variables of shrinkage were defined as negative population change, high out-migration, increased elderly population (65+), rising unemployment, low income, declining local productivity, limited social service access, and reduced quality of life. However, given data limitations across different cases and scales, the dependent variable of rural shrinkage was defined by the annual population growth rate (*r*) between two censuses, as computed with the formulation of exponential function in Eq. (1):

$$r = \frac{\text{LN}(\text{Last population} / \text{First population})}{\text{Difference of years}} * 1000 \quad (1)$$

After compiling the four datasets (socio-demographic, land-cover, facility count, and spatial network metrics), all variables were standardized as Z-scores. Then, the spatial autocorrelation was tested in ArcGIS using Global Moran's *I*. The results confirmed significant and mixed spatial dependence for key variables (e.g., Lüneburg region *wd*: *I* = -0.41; Lüneburg district population change: *I* = 0.31). Since observations were not spatially independent,

p-values from subsequent analyses were interpreted with caution (summary of Moran-*I* results: Supplementary File 9). Next, a Pearson correlation matrix was computed among four datasets in *R*. To correct for multiple testing, the Benjamini-Hochberg FDR was applied, considering only FDR-adjusted *p*-values ≤ 0.05 as significant. Significant variables were categorized, based on the absolute value of the correlation coefficient (*r*) as weak ($|0.3\text{--}0.49|$), moderate ($|0.5\text{--}0.69|$), or strong ($|0.7|$).

Once significant variables were obtained, mapping of the shrinkage typologies *k*-means clustering algorithm was applied in ArcGIS 10.3 via the Spatial Statistics Toolbox without spatial constraints and predefined seeds. When selecting variables for clustering, a hybrid expert-guided approach was used, combining statistical evidence with strategic relationships to ensure that the clusters were meaningful from a political or practical perspective. To define the number of groups (*k*), the elbow test was applied. The elbow method suggested a range of 3–4 for most datasets and was non-informative for 1. Consequently, *k* = 4 was selected, as it lied at or just after the elbow in all plots and best supported the strategic aim of distinguishing 4 meaningful urgency typologies, whereas *k* = 2–3 collapsed important nuances between settlement types. Level 1, Highly Urgent/Negative cases are marked by decreasingly positive and increasingly negative variables, which require immediate intervention. Level 2, Urgent/Vulnerable settlements indicate demographic shrinkage and aging, despite positive characteristics. Level 3, Low Urgency/Potential cases show population growth, despite negative variables. Level 4, No Urgency/Positive Cases indicate strongly positive and minor negative values (cluster centroids and settlement lists grouped by *k*-means: Supplementary File 10). This typology revealed place-based conditions and needs, identified critical cases of rural shrinkage to inform policy making and strategic spatial planning. To assess the stability of the *k*-means clustering, the algorithm was rerun 100 times for each dataset in Python with different random initializations, and the Adjusted Rand Index (ARI) was computed between runs (summarized ARI results: Supplementary File 11). Across the six cases, ARI values indicated generally high clustering stability as Mean ARI ranged from 0.62 to 0.97.

The synthesis phase was the final and defining step of the method, where the two analytical components were brought together to generate actionable strategies. It integrated: (i) the SRA–RNA prioritized factors, solutions, and weighted leverage points, which defined the essential content of interventions, with (ii) the GRA spatial network and socio-economic insights, which determined where and how resources should be allocated. This enabled qualitative evidence to inform quantitative patterns and vice versa. By specifying the why, where, how, and when of targeted resilience actions, the method linked micro-level interventions with macro-level policy recommendations (ESPON, 2021; OECD, 2025; UN-HABITAT, 2017). In doing so, the synthesis yielded a structured and evidence-informed framework that operationalized SES/CAS insights for spatial resilience planning.

4. Results

This section presents the comparative results of the study. It begins with the SRA survey results (detailed survey graphs: Supplementary File 12) and preliminary findings of RNA. Then, GRA presents the results of SNA, key variables correlated with rural shrinkage, and *k*-means groupings which define four shrinkage urgency levels.

4.1 Comparative Specified Resilience Assessment Results

4.1.1 Multi-faceted survey results

For perceived shrinkage (Survey Section 2), one third of the German respondents and half of the Turkish respondents reported ongoing rural shrinkage. The duration of perceptions diverged: 40% of Turkish respondents dated it to the past 5–10 years, while 68% of Germans believed shrinkage had persisted for 10–20 years.

As regards drivers of shrinkage (Survey Section 3), when grouped according to multiple dimensions and scales, German respondents highlighted as mezzo-scale economic (2.72) and social issues (2.53). Turkish respondents mostly attributed macro-scale economic (2.94), social (2.82), and ecological factors (2.52) to a marked difference in ecological concerns, lower in Germany and higher in Türkiye. Key factors in Germany's cases were limited diversity in the job market (3.9), lack of transportation opportunities (3.4), insufficient internet access (3.0), and regional investments (3). In Türkiye's cases, key factors were policy changes (3.7), lack of regulations on agricultural products (3.6), inadequate agricultural supports (3.6), inactive cooperatives (3.3), low regional investment (3.3), expensiveness (3.3), migration (3.2), inadequate irrigation (3), and industrial pollution (2.8). The survey also revealed some differences in the district level; for instance, Lüneburg had higher social factors (approximately 3) than Lüchow-Dannenberg (approximately 2.5), whereas compared to Babaeski, Lüleburgaz had more economic and social needs as well as pollution-related issues.

In respect of institutional roles (Section 4) for tackling shrinkage in Germany's cases, private enterprises and industrial facilities (2.96) led at the mezzo scale. While regional/state governments and institutions (2.88) were significant at macro scale, communities and personal initiatives (2.42) were key at the micro scale. Türkiye's cases indicated that ministries (2.45), central government (2.3), and regional development agencies were significant at the macro scale, with cooperatives (2.35) significant at the micro scale. District-level comparative analysis showed

that Lüchow-Dannenberg indicated a higher need for health services than Lüneburg, while cooperatives had a higher importance in Lüleburgaz, when compared to Babaeski.

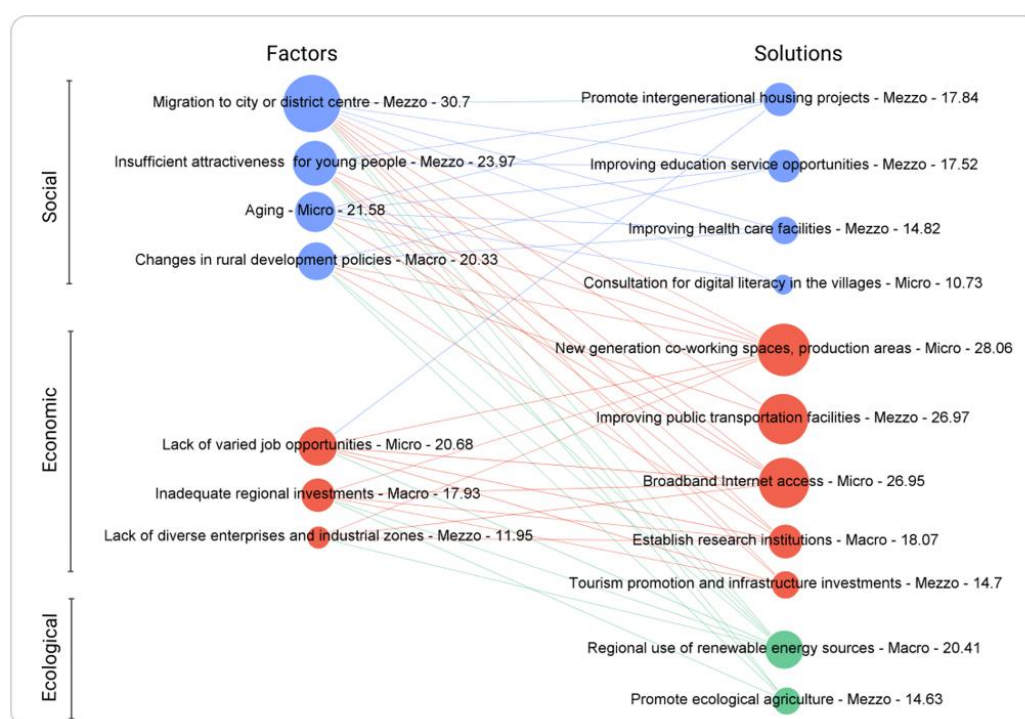
Perceived performance of institutions (Section 5) across multiple scales and dimensions were noted in Germany's cases; ecological strategies at the mezzo scale and social policies at the macro scale scored close to 5. In Türkiye's cases, social practices by provincial municipalities were rated highest (at 4.95/10), while central government policies received lower ratings around 2. Comparisons at the district level revealed that in Lüchow-Dannenberg rated local social projects (5.6) and regional strategies higher (5.11) but viewed economic policies of the central government less favorably (4.08) than Lüneburg (4.7), while Lüleburgaz rated local social projects higher (4.8) than Babaeski (2.6).

In respect of priority solutions (Section 6) grouped under multiple dimensions and scales, in Lüneburg region, social solutions (2.75) were highlighted, while in Trakya region, ecological interventions were prioritized (2.92). Both regions valued access to health (3.2) and education (3.08, 3.2) services. In Germany's cases, key solutions were increasing transport facilities at the mezzo scale (3.72), strengthening internet connectivity at the micro scale (3.67), with a notable emphasis on infrastructure and connectivity. Furthermore, Germany's cases leaned towards innovative and technological solutions like tourism promotion and infrastructure investments (2.79), regional utilization of renewable energy (2.78), and creation of new generation workplaces (2.63). Key solutions for the Trakya region included supporting agricultural production (3.85) at the macro scale, supporting cooperatives (3.68) at the micro scale, providing veterinary-agricultural engineering services in villages (3.40) at the micro scale or small production facilities in villages (3.16), and also supporting agricultural production and infrastructure.

4.1.2 Relational Network Analysis—Preliminary highlights

RNA reveals causal interconnections among prioritized factor-solution identified by survey form. Figure 6 visualizes maps of comparative RNA at regional level with weighted degrees (*wd*) of nodes. For the Germany-Lüneburg region, the most complex concern was insufficient job opportunities in both cases (Lüneburg *wd* 28, Lüchow *wd* 42), followed by migration to the city center (Lüneburg *wd* 20), aging (Lüchow *wd* 36.4), and unattractiveness for youth (*wd* 36). Common influential solutions in Lüneburg region and in other districts involve implementing novel and digital solutions (*wd* 28.06), as well as improving transport and internet facilities (*wd* 26.9).

For Trakya, highly connected complex factors included deficits in rural development policies (Lüleburgaz *wd* 39.92, Babaeski *wd* 29.3), and migration (*wd* 30-24.57), suggesting the need for diverse solutions. Among the solutions, at the regional level the highly connected solution, i.e., leverage point, was supporting cooperatives (*wd* 54.78). At the district level for Lüleburgaz, highly connected solutions were supporting cooperatives (*wd* 33.38), addressing environmental problems (*wd* 32.35), and establishing innovation centers (*wd* 27.49), whereas in Babaeski, influential solutions included the sharing economy (*wd* 36.2), innovation centers (*wd* 33.07), and supporting commercial small and medium enterprises (*wd* 31.77).



(a)

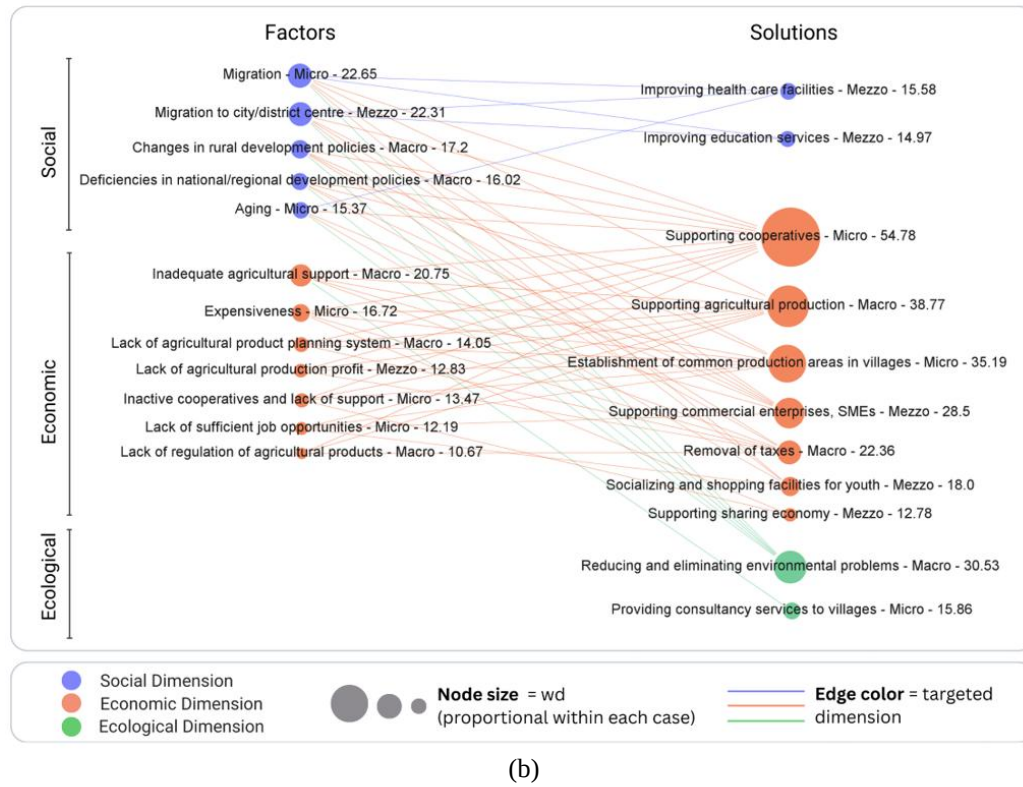


Figure 6. Factor-solution relational network maps. (a) Lüneburg Region and (b) Trakya Region
 Note: Node size is proportional to weighted degree (wd) within each panel; edge color indicates targeted dimension. Nodes with $wd \leq 10$ are omitted for readability.

4.2 Comparative General Resilience Assessment Results

4.2.1 Results of Spatial Network Analysis

Figure 7 shows transport connectivity networks across regions and districts; while node size represents wd , colors indicate modularity sub-clusters (Blondel et al., 2008). SNA at the regional scale revealed that in the Lüneburg region, Lüneburg district ranked second with wd 28 but Lüchow-Danneberg was the 2nd lowest, at wd 14. In the Trakya region, Lüleburgaz had the highest wd 36, followed by Babaeski wd 21. Compared to Türkiye's cases, Lüneburg's districts exhibited close wd values and had more even distribution.

Conversely, the central districts in Trakya, notably those located near Trans-European Motorway (TEM), exhibited higher wd values, while the peripheral districts had low wd values. At the micro scale, Türkiye's villages showed a polycentric structure, while Lüneburg had a single center and Lüchow-Danneberg had dual centers with Lüchow and Dannenberg.

In terms of modularity sub-clusters, the Trakya region had four, while Lüneburg region had two. At the district level, Lüleburgaz and Lüneburg each contained four sub-clusters, whereas Babaeski and Lüchow-Danneberg comprised five sub-clusters. This spatial analysis highlights patterns of settlement connectivity and identifies sub-clusters and sub-centers, which are pivotal in the synthesis phase.

4.2.2 Results of correlation analysis

Variables demonstrating statistically significant correlations (FDR-adjusted $q \leq 0.05$) with a coefficient strength of $|r| \geq 0.5$ with population growth and general development attributes are consolidated in Table 1 and then explained below.

Positively correlated factors on population growth: across land-cover types, urban and industrial areas showed a strongly positive correlation with population growth or population size at all scales, while in the Lüneburg region of Germany, only urban land cover was significant. In terms of facility types, social, commercial, and transportation facilities correlated with population growth in all examples. In the Turkish examples, more diverse facilities (social, recreational, commercial, transportation, and religious) demonstrated a positive correlation with population growth. In terms of network metrics, settlements with high wd and betweenness centrality generally experienced population growth; closeness centrality also illustrated a positive relationship in some cases.

Among the negative correlations of development, eccentricity metric and distance to center showed a negative correlation with general development attributes and population growth, except in Lüleburgaz. Furthermore, a high

proportion of agricultural areas in the Lüneburg and Lüneburgaz districts had a negative correlation with population growth; a high proportion of forest areas in the Babaeski district had a significant correlation with elderly population.

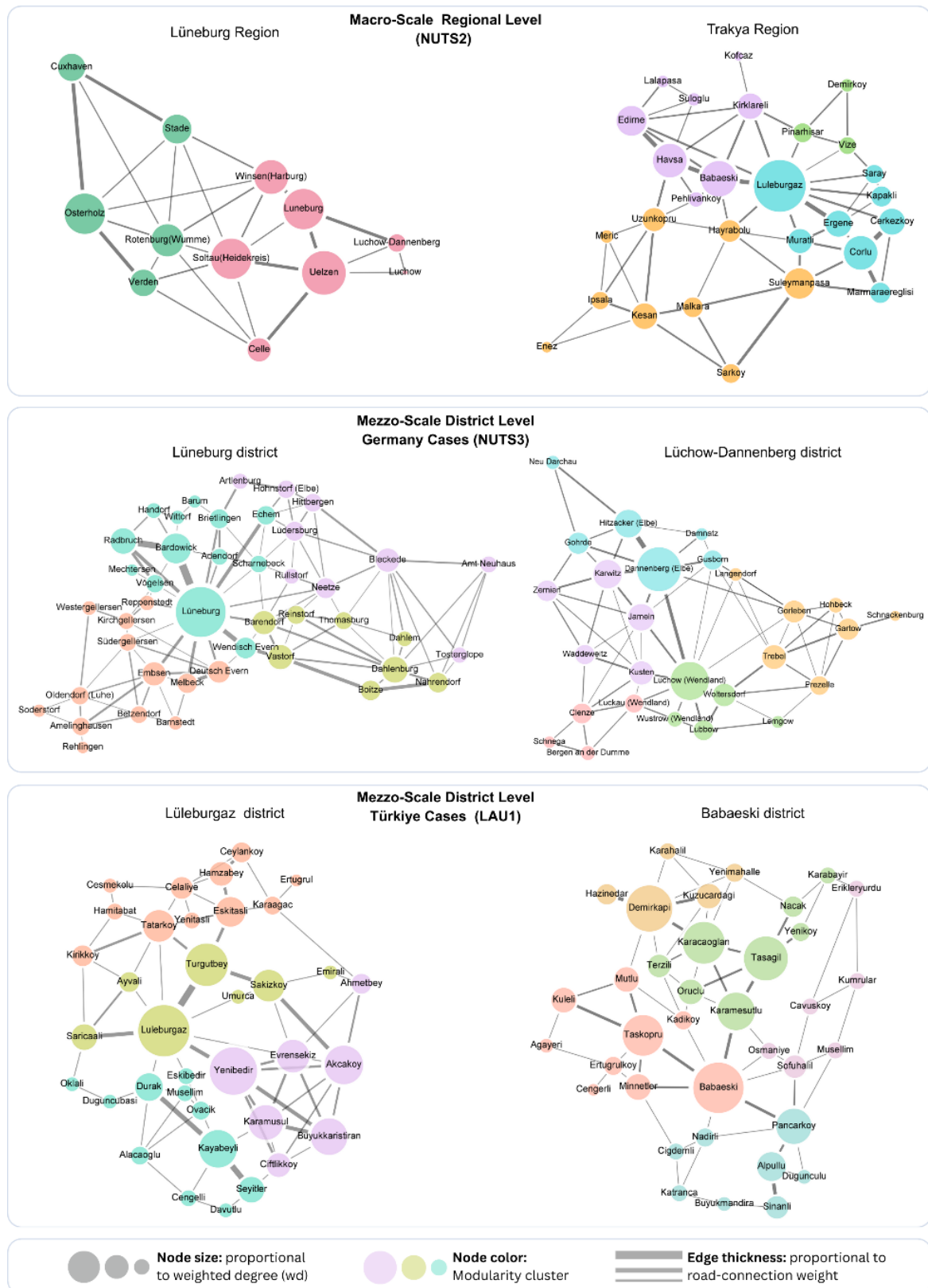


Figure 7. Spatial network maps of the case studies

Note: Node size is proportional to weighted degree (wd) and scaled within each panel; node color indicates modularity-based sub-clusters; edge thickness is proportional to road-connection weight.

Table 1. Variables with significant False Discovery Rate (FDR)-adjusted correlations ($q \leq 0.05$) with population change rate or population size across all cases

Macro-Scale Regions		Mezzo-Scale Germany Districts		Mezzo-Scale Türkiye Districts	
Lüneburg	Trakya	Lüneburg	Lüchow-Dannenberg	Lüleburgaz	Babaeski
Positively Related Variables					
Land-Cover Types vs. Population Change Rate or Population Size					
0.54 Urban LC	0.55 Urban LC	All types of LC vs. Pop. 2018	0.94 Urban LC vs. Pop. 2018	0.7 Urban LC	0.59 Urban LC
	0.66 Industrial LC		0.8 Industrial LC vs. Pop. 2018	0.67 Industrial LC	0.51 Industrial LC
Facility Types vs. Population Change Rate or Population Size					
0.66 Social F.	0.81 Com. F.	All types of Fac. vs. Pop. 2018	All types of Fac. vs. Pop. 2018	0.71 Com. F.	0.57 Social F.
0.53 Com. F.	0.74 Social F.			0.71 Rec. F.	0.54 Com. F.
	0.73 Transport F.			0.68 Religious F.	0.54 Transport F.
	0.59 Rec. F.			0.67 Social F.	0.46 Religious F.
				0.65 Environ. F.	
				0.56 Transport F.	
Network Metrics vs. Population Change Rate or Population Size					
	0.59 <i>Wd</i>	0.87 Betw. Cen. vs. Pop. 2018	0.75 <i>Wd</i> vs. Pop. 2018	0.58 Betw. Cen.	0.55 Betw. Cen.
		0.79 <i>Wd</i> vs. Pop. 2018	0.68 Betw. Cen. vs. Pop. 2018	0.5 <i>Wd</i>	
				0.5 Clos. Cen.	
Combination of Negatively Impacting Variables on General Development					
		-0.54 Dist. to Center	0.57 Distance to Lüchow vs. Elder	-0.4 Dist. to Center	0.59 Forest LC vs. Elder
		-0.44 Agri LC		-0.57 Agri LC	-0.55 Dist. to Center

Note: LC = Land-cover; F. = Facilities; Com. = Commercial, *Wd* = Weighted degree; Betw. Cen. = Betweenness centrality; Clos. Cen. = Closeness centrality; Elder = Elderly population share (65+); Dist. to Center = Settlement distance from district center.

To sum up, population growth positively correlated with urban and industrial land cover, transportation, social facilities, and certain network metrics (*wd* and betweenness centrality). Conversely, land-cover types like agriculture and forest, alongside distance to center (high eccentricity) were associated with rural shrinkage. These findings underscore the associations among socio-demographic trends, land-cover types, facility assets, and transport connectivity, that will guide the development of strategic interventions in the synthesis (Section 5).

4.2.3 Results of *k*-means analysis

The *k*-means analysis condensed extensive demographic, land-cover, facility and spatial network data into a four-level urgency map, highlighting issues for the focus of resilience measures. Section 5 illustrates this four-level urgency classification (color fills) with the network sub-clusters (polygon outlines) across all three scales. Empirical results for each case study are explained below. Section 5 revisits these spatial patterns, integrating them with the SRA results, to develop concrete resilience strategies.

Among the macro-scale regional results, in the Lüneburg region (11 districts): Level 1 ($n = 2$) ranked lowest on facility counts and network metrics (excluding *wd*) and highest on both unemployment and elderly population. Level 2 ($n = 1$) had a high number of commercial facilities and urban land cover, but struggled with low network metrics, population decline, and high elderly share. Level 3 ($n = 3$) indicated that districts' variables had median values. Level 4 ($n = 4$) showed the highest numbers of facilities and network metrics, high population growth, minimal elderly population, and low eccentricity metric.

In the Trakya region (27 districts): Level 1 districts ($n = 11$) located in peripheries with the lowest network metrics, fewest facilities, and the highest share of elderly population. Level 2 ($n = 11$) was characterized by significant elderly population but median values for all other variables. Level 3 ($n = 4$) displayed higher network metrics and more facilities, plus the lowest elderly share and stably lower population growth. Level 4 ($n = 1$) was solely Tekirdağ center district, marked by the highest population growth, lower elderly ratios, and higher network metrics with other influential variables.

Among the mezzo-scale district and micro-scale settlement results, in the Lüneburg district (43 settlements): Level 1 ($n = 5$) faced the highest population decline and elderly population. Level 2 ($n = 1$), as the district center, experienced the second-highest population decline and elderly rates. Level 3 ($n = 21$) had lower facilities and network metrics but showed significant population growth. Level 4 ($n = 14$) had maximum facility numbers, network metrics, population growth, and minimum elderly population.

In the Lüchow-Dannenberg (27 settlements): Level 1 ($n = 7$) settlements had the highest population decline and elderly population. Level 2 ($n = 13$) suggested second-highest population decline and elderly population. Level 3 ($n = 5$) had the lowest facilities and network metrics but second-highest population growth. Level 4 ($n = 2$), as the two district centers, had the maximum facility numbers, network metrics, population growth, and the lowest elderly population.

In the Lüleburgaz district (35 settlements): Level 1 ($n = 17$) suggested villages located in peripheries and had the lowest population growth, facilities, and network metrics. Level 2 ($n = 11$) had better network metrics but low population growth and industrial areas. Level 3 ($n = 5$) showed high population change and facilities but lower network metrics. Level 4 ($n = 1$) implied that the district center excelled in all metrics.

In the Babaeski district (36 settlements): Level 1 ($n = 17$) highlighted villages located in peripheries, noted for highest elderly rates, declining population, and low centrality metric. Level 2 ($n = 13$) indicated high network metrics, though struggling with population decline and low industrial zones. Level 3 ($n = 5$) had lower network metrics, yet high industrial areas and population growth. Level 4 ($n = 1$) indicated that the district center had positive status for all variables.

As a result, similar trends emerged across all cases; low-urgency areas had stronger network centrality and richer facilities, while high-urgency areas were characterized by their peripheral locations and weak linkages. These quantitative contrasts provide the empirical basis for the targeted interventions developed in Section 5.

5. Synthesis of Specified Resilience Assessment and General Resilience Assessment Findings and Strategic Interventions

The final phase of the methodology synthesized the analytical outputs of the SRA–RNA, prioritized needs, and leverage points, with the GRA spatial network metrics and socio-economic shrinkage urgency typologies to derive actionable strategies. To translate these insights into concrete and defensible policy tools, two complementary rules were defined to prioritize interventions, distinguished between identifying capacity nodes and addressing vulnerability linkages. Figure 8 outlines the operational steps required to reproduce the workflow and formulate strategies, while Figure 9 presents the resulting shrinkage urgency levels, spatial network sub-clusters, and suggested strategic service hubs and linkages.

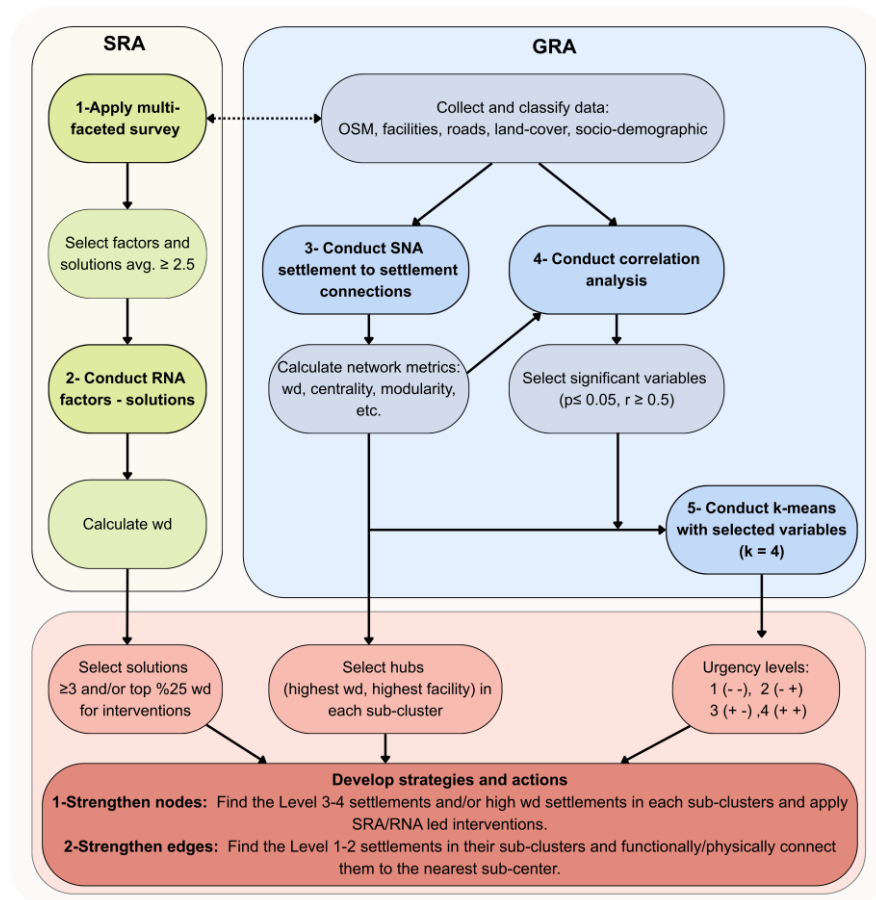


Figure 8. Operational steps of the method

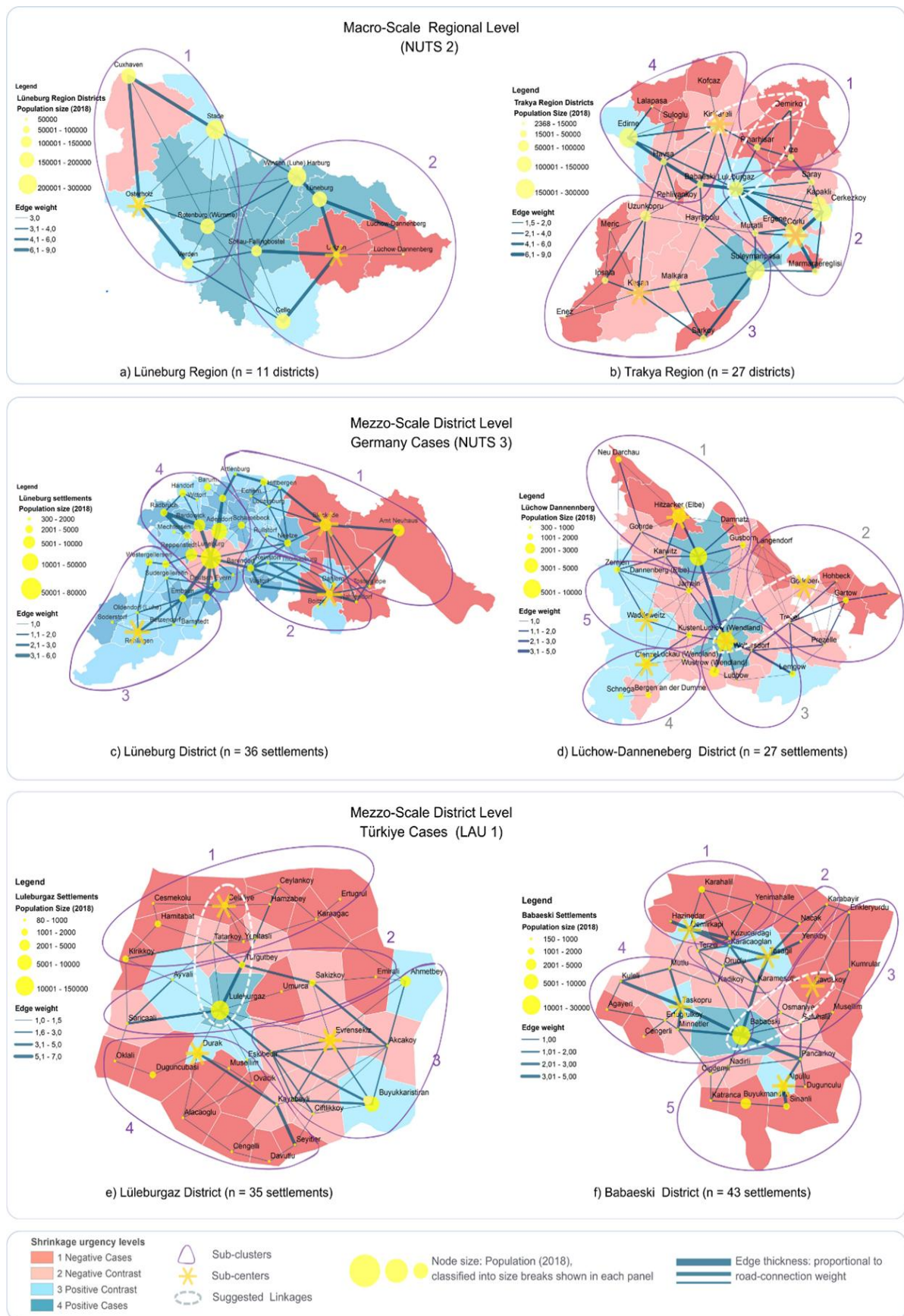


Figure 9. Maps of synthesized network and k -means analysis

Note: Node size represents 2018 population and is scaled within each case with the class breaks as shown, not intended for direct cross-panel comparison.

Strategy 1, strengthening nodes, aims to identify stable settlements that could function as primary service providers for their respective sub-clusters. A settlement is designated as a sub-center for investment if it meets the following criteria:

- Spatial centrality: The node has the highest weighted degree (top 1 *wd*) within its spatial network sub-cluster, indicating high physical accessibility.
- Capacity/stability: The node is classified as Level 3 (potential) or Level 4 (positive) in the *k*-means cluster analysis, indicating relative demographic stability and economic capacity.
- If no capacity but strong accessibility: Where no node in the sub-cluster falls into Level 3–4, a highly accessible node (high *wd*) can still be selected and strategically strengthened through the SRA-derived interventions.
- If no strong accessibility but clear capacity: Conversely, if no node has a particularly high *wd* but one or more nodes show strong capacity (Level 3–4), these capacity nodes are selected and their roles reinforced by improving linkages and facilities.
- Defining the intervention content: Once a node is selected, the specific intervention package is derived from SRA–RNA by combining solutions with a Survey Average > 3 and/or those at the top 25% *wd* ($\geq 75^{\text{th}}$ percentile) of RNA.

Strategy 2, strengthening linkages, aims to mitigate isolation and connect at risk areas to the nearest sub-center. A new or upgraded functional/relational linkage is identified as a priority link if it meets the following criteria:

- It originates from a Level 1 (highly urgent) or Level 2 (urgent/vulnerable) settlement.
- It connects this at-risk settlement to its nearest sub-center (as defined by Rule 1), ensuring access to key services.
- Linking upgrade can include raising edge class one step (e.g., from tertiary to secondary) or adding scheduled transit (≥ 6 trips/day) or, establishing shared service (mobile clinic and weekly admin desk) if road upgrade is infeasible.

These strategies underpin policies and actions that ensure efficient service delivery and foster integrated regional/spatial development in response to rural shrinkage. Section 5.1 sets out macro-scale and long-term policies for the Lüneburg and Trakya regions, while 5.2 details mezzo-scale mid-term plans and micro-scale short-term actions for each district and settlement levels. Together, a hybrid and replicable resilience assessment method was operationalized to guide strategic decision making, thereby strengthening resilience and supporting sustainable development.

5.1 Macro-Scale Strategies (Regional Level)

In the Lüneburg region (Figure 9a), high-urgency districts are spatially concentrated on the periphery, contrasting with positive cases clustered near the regional center. Applying Rule 1 (Strengthening Nodes), the analysis identified the highly connected and stable nodes of Osterholz (Cluster 1) and Uelzen (Cluster 2) as sub-centers. These were designated as logistic sub-centers, capable of servicing the wider region. Rule 2 (Strengthening Linkages) subsequently targeted the peripheral and shrinking areas to establish robust linkages to these hubs. The operational content for these sub-centers was defined by the SRA priorities, focusing on enhancing public transport (3.72) and broadband access (3.67) to mitigate isolation, alongside reinforcing health and educational services (3.16) to address the specific needs of the elderly population and limited job markets.

The Trakya region (Figure 9b) displays distinct spatial disparities, with Cluster 1 exhibiting the weakest socio-economic conditions. Rule 1 identified a network of highly connected sub-centers to anchor regional development: Ergene (Cluster 2), Edirne (Cluster 4), and the highly connected Süleymanpaşa and Keşan (Cluster 3). For the vulnerable Cluster 1, Rule 2 indicated the creation of linkages to the nearest stable hubs, specifically Lüleburgaz or Kırklareli, to ensure targeted intervention delivery. Aligned with stakeholder inputs, long-term policies for these nodes could focus on increasing agricultural support (3.85), promoting sharing economy (3.25), and addressing environmental issues (3.1) to build regional resilience.

5.2 Mezzo-Scale Strategies (District Level)

At the district level, the rules generated settlement-specific action plans. In Lüneburg district (Germany) (Figure 9c), urgent Level 1 settlements were concentrated in the eastern areas (Clusters 1 & 2) and notably the district center was Level 2, while the western Hamburg commuter zone demonstrated strong capacity (Level 3/4). According to the operational selection rules, Bleckede (Cluster 1) and Dahlenburg (Cluster 4) could be the sub-centers due to their high *wd*. To empower these sub-clusters, SRA-identified leverage points such as diverse job opportunities (4), transportation (3.92), and broadband (3.55) to reverse demographic decline could be utilized.

Lüchow-Dannenberg district (Germany) (Figure 9d) presented a challenging profile where most settlements were vulnerable Level 2, with high-urgency Level 1 cases like Hitzacker concentrated in the north, while Level 3 settlements located in the southern part. Rule 1 identified these southern Level 3 settlements as the sole viable sub-

centers for service delivery in their sub-clusters. Rule 1 also highlighted the high *wd* hubs like Hitzacker and Gorleben, possible sub-centers with the empowerment of their capacities. For Cluster 2, which had no positive cases, Rule 2 identified a linkage with Lüchow center. The strategic output therefore focused on reinforcing southern nodes with healthcare facilities (3.15) to serve the aging population, while simultaneously upgrading transport infrastructure to the north to address youth unemployment (3.85) and digital isolation (3.77).

In the Lüleburgaz district (Figure 9e); Clusters 1 and 4 contain most of the Level 1 villages. Settlements like Tatarköy or Celaliye (Cluster 1), Sakızköy (Cluster 2), Evrensekiz (Cluster 3), Alacaoğlu or Durak (Cluster 4), could act as service hubs with strong network metrics. Mid-term plans for these sub-centers could include supporting cooperatives (3.83), provide rural consultancy services (3.83), and establish shared production areas (3.33). Level 3 settlements, noted for their high industrial facilities and population growth, such as Ayvalı (Cluster 2), Ahmetbey (Cluster 3), and Durak (Cluster 4), were suited to be logistic centers and leverage measures such as supporting commercial enterprises, sharing economy (3.5), improving health and educational services (3.38), alongside mitigating industrial pollution (3.2) to improve capacities of these hubs. Applying Rule 2, Level 1-2 settlements could connect to these Level 3 or Level 4 settlements after improved conditions.

In the Babaeski district (Figure 9f), Level 1 and 2 settlements were located predominantly on the periphery of the district, thus indicating immediate resilience solutions. Level 3 settlements such as Demirkapı (Cluster 1), Taşağıl (Cluster 2), Alpullu (Clusters 5), and Taşköprü (Cluster 4), rich in industrial areas, could function as business and service sub-centers within their sub-clusters and their connections with Levels 1 and 2 could be improved. Despite their positive conditions, Level 3 sub-centers required aging-related policies. Rule 2 specifically targeted Cluster 3, which lacked a positive center, mandating a priority link to connect its settlements to the main district center. Presenting innovative entrepreneurial centers in Level 3 settlements could foster job diversity (3.3), improve health (3.29) and education (3.13) services. Villages with high node degrees (Demirkapı, Taşağıl, Sofuhalil, Taşköprü, and Pancarköy) are suitable for micro-scale and short-term interventions as leverage points like consultancy services (3.5), cooperative support (3.5), and shared production facilities (3).

6. Discussion

Resilience research within the SES domain has long employed CAS theory to unravel complex systems. Yet in spatial planning, resilience assessments often remain fragmented though the focus is on single systems or issues without fully encompassing CAS methodologies, separated quantitative and qualitative methods, or concentrated mainly on fast variables. As a result, a notable gap exists in comprehensive and hybrid spatial resilience assessment, particularly for slow variables.

This study addressed this gap by developing a multi-faceted and network-based hybrid method that integrated SES and CAS principles into spatial planning. Rural shrinkage was selected as the empirical test case, due to its slow-variable influence on the SES functionality and its close linkage to pressures of global urbanization and challenges of sustainable development.

The proposed multi-faceted and network-based hybrid resilience assessment method combined five analytical steps within a unified framework for strategic spatial planning. In the SRA component, (i) a multi-faceted survey identified and prioritized place-based challenges, solutions, and institutional roles; and (ii) RNA mapped causal interdependencies between selected risks and solutions to reveal complex problem constellations and leverage points. In the GRA component, (iii) SNA identified sub-clusters and sub-centers and local/global network metrics; (iv) correlation analysis revealed key socio-economic and spatial drivers of shrinkage associated with population change; and (v) *k*-means clustering classified regions and districts into four shrinkage urgency levels, creating a shrinkage typology. Synthesizing the outputs from SRA and GRA enabled a place-based and data-driven decision-support tool that consolidates qualitative local perspectives with quantitative statistical and spatial analytics.

The comparative application of the method in paired regions of Germany and Türkiye highlighted distinct place-based needs and characteristics, to be distinguished between the contexts of developed and developing countries. In Germany (Lüneburg region), a more balanced mix of positive and negative patterns was observed. For cases in Germany, SRA-RNA analysis identified mezzo and micro-level measures such as diversifying employment, enhancing social services for aging populations, and investing in innovative infrastructure like coworking spaces, broadband connectivity, and renewable energy. Cases in Türkiye (Trakya region) exhibited predominantly negative patterns driven by macro-scale factors including agricultural policy changes and ecological problems linked to industrial activities. Relational analysis underscored multi-level support, focusing on promoting cooperatives, boosting agricultural production, and enhancing social services.

Across both cases, common spatial findings highlighted that growth was associated with urban and industrial land cover ($r = 0.54$) and well-connected settlements (*wd*; r up to 0.79). In contrast, shrinkage was associated with higher shares of forest/agricultural areas ($r \approx 0.5$) and more peripheral and low-centrality settlements ($r \approx 0.55$), which also correlated with higher elderly ratios ($r \approx 0.58$). These insights underscore the vulnerability of remote rural areas and highlight the importance of strong linkages and proximity to centers as key factors to mitigate rural shrinkage. Therefore, understanding distinct local needs and developing measures informed by spatial network

metrics, proximity to centers, and land-cover types are critical determinants of rural resilience against shrinkage.

Synthesizing the SRA and GRA outputs generated two operational selection rules. Rule 1 (Strengthening nodes): prioritize highly connected and/or stable nodes (Level 3/4) as service hubs and empower them with SRA-derived interventions. Rule 2 (Strengthening linkages): connect shrinking and peripheral settlements (Level 1/2) to these sub-centers to improve resource allocation and service delivery.

Consequently, these insights point to five system-wide principles to enhance resilience strategies:

- Dynamic monitoring of place-based needs and leverage points;
- Mapping spatial sub-clusters and designating sub-centers;
- Prioritizing targeted investments in sub-centers to optimize resource allocation;
- Linking peripheral and shrinking settlements to at least one sub-center or low-urgency settlement to improve service delivery and integration; and
- Strengthening functional/relational/physical linkages among settlements to enhance overall connectivity and resilience.

The method contributes to spatial resilience planning in five specific ways, hence distinguishing it from existing frameworks:

- (1) Operationalizing complexity (SES/CAS): It integrates complex system perspectives into a multi-faceted operational workflow that captures interactions across multiple levels, scales, dimensions, and temporalities. The network-based approach reveals relational interdependencies among diverse factors and solutions via RNA and maps spatial linkages via SNA to strategically identify the hubs and sub-clusters that underpin complex system attributes.
- (2) Addressing slow-variable disturbances: Most urban resilience frameworks provide toolkits and metrics focused on fast variables, shocks, and infrastructure resilience (Cardoso et al., 2020; CDRI, 2025; UNDRR & CDRI, 2023). This study addressed a critical gap by testing the developed method specifically on a slow-variable disturbance, i.e., socio-demographic shrinkage. The method demonstrated how resilience planning could be adapted for long-term systemic stressors often missed by shock-focused assessments.
- (3) Bridging policy strategies and operational actions: The primary novelty lies in bridging the critical gap between strategic policy orientation and operational action. Existing policy frameworks such as ESPON (2017), OECD (2025) and UN-HABITAT (2017) provide essential strategic orientation (e.g., enhance service delivery) but often do not specify reproducible mechanisms for deciding where and how to implement these goals at the settlement level. This study addressed this by directly operationalizing policy calls for place-based, data-driven, and network-oriented governance and spatial interventions. By identifying sub-centers and linkages, it provided the micro-level operational tool required to achieve macro-level smart shrinkage strategies and enhance urban-rural linkages.
- (4) Scalability and generalizability: Although this study focused on rural shrinkage, the multi-faceted and networked approach of the method provided a standardized workflow for assessing the complexity and resilience of other issues. RNA supports the assessment of interdependence among multiple risks and cascading effects; SNA could be used for infrastructure network assessment via hubs and sub-clusters; and GRA contributes by grouping urgency levels based on spatial and socio-demographic attributes.
- (5) Embedding in strategic spatial planning practice: Within the strategic spatial planning processes, the SRA could be integrated into focus groups and desk meetings to identify leverage points and priority interventions. GRA outputs could also be embedded into existing planning tools. They could (i) inform regional plan decisions by the designation of sub-centers and sub-clusters; (ii) support periodic monitoring by regularly re-running indicators and clustering as an early-warning tool; and (iii) guide rural development programs by adding spatial layer of shrinkage-urgency classes and hub-periphery relations.

While this study presented an integrated and hybrid framework, its findings were subject to several important limitations and should be interpreted as guidance for future research. At a general level, the study should be framed as a methodological proof of concept. The primary aim is not to produce statistically generalizable and population-level findings but rather to develop and test the feasibility of integrating different analytical methods (SRA-RNA, GRA-SNA). The following limitations are a consequence of the scope of this pilot study:

- The SRA survey included a small and non-probability participant pool, and RNA was created by a single researcher. This introduced risks of selection bias and researcher bias. Accordingly, the identified priorities and leverage points (factors/solutions) should be treated as exploratory and case-specific rather than generalizable findings.
- Given mixed statistical indications of elbow test, as stated in the methodology, the choice of $k = 4$ for k -means clustering reflected a strategic decision for policy purposes. Another researcher might reasonably select a different number of clusters and obtain alternative typologies.
- The variable set used for clustering was selected through a hybrid and expert-guided process combining statistical correlation with politically important variables. Consequently, the cluster solution was not a purely data-driven discovery, but a guided classification shaped by expert judgement. This selection could be revisited and refined according to different priorities.

- Relational and spatial adjacency matrices created manually were labor-intensive and produced a static representation at a single point in time. This approach did not easily accommodate real-world changes (e.g., new roads and infrastructure upgrades) and limited the dynamism of the model.

As a result of these limitations (manual processes, subjective choices, and small samples), this method is not yet ready for automation. As mentioned above, this method is proof of concept that requires significant improvement before it could be scaled for rapid or dynamic application.

Based on these limitations, future studies should broaden the SRA participation to a larger and more diverse stakeholder pool to enhance generalizability. In particular, the RNA structure should be co-validated by multiple independent experts to reduce researcher bias. Future work could also expand the range of networks to be considered (e.g., transport, digital, ecological, and service networks) and quantify flows on them (e.g., commuting patterns, service usage, and mobility flows), to enable a more dynamic and functional assessment of system connectivity and resilience. To move beyond a static model, subsequent research should focus on automating both relational and spatial network construction and, ideally, linking them to live statistical databases. This would support dynamic monitoring and broader application, which are the longer-term ambitions of this research. Moreover, while the framework identified the starting point to intervene, detailed cost, budget, and feasibility assessments for specific interventions were beyond the scope of this study and should be developed in future work by responsible agencies.

7. Conclusions

This study developed and tested a multi-faceted, network-based hybrid resilience assessment method for spatial planning by integrating SRA and GRA through SES and CAS perspectives. Combining a multi-faceted matrix, survey-based diagnosis, RNA, SNA, correlation analysis, and grouping analysis, the method translates complex systems thinking into an applied framework for examining slow-variable disturbances. It enables the analysis of system conditions, interdependencies, governance roles, intervention priorities, and spatial patterns across multiple scales, dimensions, institutional levels and time horizons. The synthesis of qualitative and quantitative findings generated two operational planning directions: strengthening sub-centers through targeted investments and linking shrinking settlements to these centers to improve access, service provision, and regional integration.

Applied comparatively to rural shrinkage in Germany and Türkiye, the method identified place-based needs that differ according to developmental status, while also showing that peripheral, weakly connected settlements face greater shrinkage risk, whereas better-connected centers display stronger resilience capacities.

These findings should be interpreted in light of the study's limitations, including small survey samples, case-specific data harmonization constraints, and partly manual network construction. The method should therefore be understood as a transferable proof of concept, with future research needed to broaden participation, strengthen validation, and further automate the workflow.

In line with international policy calls, the study demonstrated that coupling multi-faceted place-based insights with quantitative network analysis can translate resilience theory into actionable and spatially explicit planning strategies. The method thus offers a holistic decision-support framework for targeted interventions toward more resilient and sustainable spatial development pathways.

Author Contributions

Conceptualization, B.A.; methodology, B.A.; validation, B.A. and A.T.; formal analysis, B.A.; investigation, B.A.; resources, B.A.; data curation, B.A.; writing—original draft preparation, B.A.; writing—review and editing, B.A. and A.T.; visualization, B.A.; supervision, A.T.; funding acquisition, B.A. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data and scripts used to support the research findings are available in the repository file: <https://github.com/BilgeAyd/multi-faceted-network-based-resilience-assessment>

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Conflicts of Interest

The authors declare no conflicts of interest.

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